Project Overview

This project is designed to detect unsafe tilt angles of a portable device or cargo box using an STM32 microcontroller and MPU6050 sensor.

The system continuously monitors the orientation of the device. When the tilt angle exceeds a predefined safe limit:

- An LCD warning indicator turns ON
- A warning message is transmitted through UART (Serial Monitor)

This project demonstrates:

- Sensor interfacing using I2C
- Real-time tilt monitoring
- UART communication
- Embedded warning system design

Objectives

The objectives of this project are:

- Interface MPU6050 with STM32
- Read accelerometer values in real time
- Calculate tilt angle
- Detect unsafe orientation
- Trigger visual using LCD
- Send warning messages through UART

Components Required

Component	Quantity
STM32 Nucleo Board	1
MPU6050 Sensor	1
LCD	1
220Ω Resistor	1
Jumper Wires	Several

System Working Principle

The MPU6050 continuously measures acceleration values along:

- X-axis
- Y-axis
- Z-axis

The STM32 processes these values to determine the tilt angle.

If the angle exceeds the critical limit (example: 30°):

- LCD turns ON
- UART displays warning:
  `"WARNING: Device Tilt Detected!"`

Otherwise:

- LCD remains OFF
- UART displays:
  `"System Stable"`

Block Diagram

```
MPU6050 Sensor
      |
      | I2C Communication
      |
```

```
        STM32 Microcontroller
          |           |
          |           |
         LCD        UART Monitor
       Warning      Warning Message
```

Circuit Connections

MPU6050 to STM32

MPU6050 Pin	STM32 Pin
VCC	3.3V
GND	GND
SDA	PB7
SCL	PB6

LED Connection

LCD Pin	STM32 Pin
Anode (+)	PA5
Cathode (-)	GND through 220Ω resistor

Main Program Flow

```
Start
 ↓
Initialize MPU6050
 ↓
Read accelerometer data
 ↓
Calculate tilt angle
 ↓
Is angle > threshold?
 ↓         ↓
```

YES NO
↓ ↓
LCD ON LCD OFF
Send Send
Warning Stable
Message Message
↓
Repeat forever

Example UART Output

Tilt Angle: 12.5
System Stable

Tilt Angle: 18.2
System Stable

Tilt Angle: 35.7
WARNING: Device Tilt Detected!

Expected Simulation Result

Normal Position

- LCD OFF
- UART shows “System Stable”

Tilted Position

- LCD ON
- UART warning displayed

In Wokwi:

- Click the MPU6050
- Change X/Y tilt values
- Observe LCD and Serial Monitor response

Applications

This system can be used in:

- Cargo shipment monitoring
- Medical equipment transport
- Laptop protection systems
- Industrial equipment monitoring
- Precision instrument handling

Advantages

- Low cost
- Real-time monitoring
- Simple implementation
- Reliable warning system
- Easy to expand

Conclusion

The Tilt Warning System successfully detects unsafe orientation using the MPU6050 sensor and STM32 microcontroller.

The system provides:

- Visual warning through LED
- Serial warning through UART

This project demonstrates practical implementation of embedded sensing, real-time monitoring, and warning systems commonly used in industrial and commercial applications.